

Lesson 9: Coupling

Geometry controls shared changing flux

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

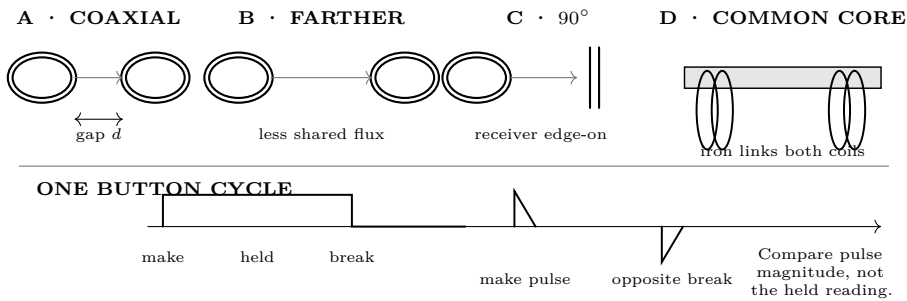
What you need

- Two insulated coils
- 2 AA cells and pushbutton
- Multimeter with peak/min-max mode

- Removable iron core
- Ruler, protractor, graph paper
- Tape and nonmetallic supports

Predict first

Rank these changes before testing: halve the gap, rotate the receiver 90°, insert a common iron core, or double the meter-lead length. Which should change the received pulse most? Why?



Investigate

1. Tape both coils coaxial at $d = 0$ cm. Fix the battery, button, meter range, lead routing, and one operator. Mark coil faces and lead polarity.
2. Make five brief, equally timed button presses. Record the largest absolute receiver voltage for each make pulse; use their median. Also record the break-pulse sign. Stop if a coil or cell warms.
3. Repeat at $d = 1, 2, 4, 8$ cm. Move only the receiver along the common axis. Do not stretch or reroute leads.
4. Return to $d = 2$ cm. Repeat at $\theta = 0^\circ, 30^\circ, 60^\circ, 90^\circ$, rotating about the receiver's center.
5. Repeat the distance series with the removable iron core linking both coils. Remove it between trials; check that the zero-gap baseline returns.

Distance × core measurement matrix				
d (cm)	Air: median $ V $	Core: median $ V $	Core/air	What stayed fixed?
0				
1				
2				
4				
8				

Orientation matrix at $d = 2$ cm				
θ	median $ V $	fraction of 0°	projected-area prediction	observed pulse sign
0°		1.00	maximum	
30°				
60°				
90°			near minimum	

Explain from the evidence

The transmitter's changing current makes changing magnetic flux. The receiver responds only to the portion linking its turns:

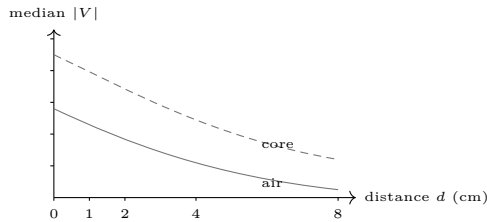
$$V_{\text{RX}} \approx -M \frac{\Delta I_{\text{TX}}}{\Delta t}, \quad M = k\sqrt{L_{\text{TX}}L_{\text{RX}}}.$$

Geometry and the core change the coupling factor k ; the button edge changes $\Delta I/\Delta t$. A steady held current has flux but little change, so the receiver pulse falls toward zero. Releasing the button reverses the flux change and therefore the pulse polarity. This is induction, not wireless creation of energy.

Interrogate the evidence	
1. Which single trial is your baseline?	8. Did the core multiply every distance by the same factor?
2. Did five pulses agree closely enough to compare?	9. Which result supports shared flux rather than lead length?
3. What happened when d doubled?	10. Why is the held reading not the comparison value?
4. Is the distance trend linear? What evidence says so?	11. Why do make and break pulses have opposite signs?
5. Why must the coil centers stay on one axis?	12. Which variable changes M ? Which changes $\Delta I/\Delta t$?
6. At which angle did the response halve?	13. What observation would falsify your geometry claim?
7. Was 90° exactly zero? Why might it not be?	14. Where is measurement resolution limiting the conclusion?

Change one thing

Graph the relationships. Plot median $|V|$ against distance for air and core on the same axes. Add a second graph of normalized response $|V_\theta|/|V_0|$ against angle. Plot measured points first; connect only as a visual guide. Label units, baseline, meter range, and any below-resolution point. Do the curves justify “farther is weaker” and “edge-on is least,” or only suggest them?



Fault isolation. If the receiver pulse is missing or erratic, change one cause at a time.

Symptom	Discriminating check
No make or break pulse	Short the meter probes together, restore them, then verify meter range and receiver continuity.
Pulse only when leads move	Tape leads; press without touching either coil.
Every trial varies	Use one operator and five-pulse median; inspect button.
90° remains large	Check centers, stray metal, and lead loops.
Core seems weaker	Return to air baseline; inspect whether core actually passes through both coil fields.

FINAL PROOF

Blind the receiver configuration behind a sheet. A partner chooses two of these states: coaxial/near, coaxial/far, orthogonal, or common-core. From five pulses per state, predict which geometry is which and justify the identification from the distance, orientation, and core graphs. Then swap roles. The claim passes only if both identifications are correct, the baseline returns within the trial spread, and make/break polarity reverses. Record any failure as evidence and isolate it before repeating.

Proof record			
Prediction	_____	Controlled variables	_____
Strongest evidence	_____	Uncertainty	_____
Blind state A	_____	Blind state B	_____
Claim + because	_____		

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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